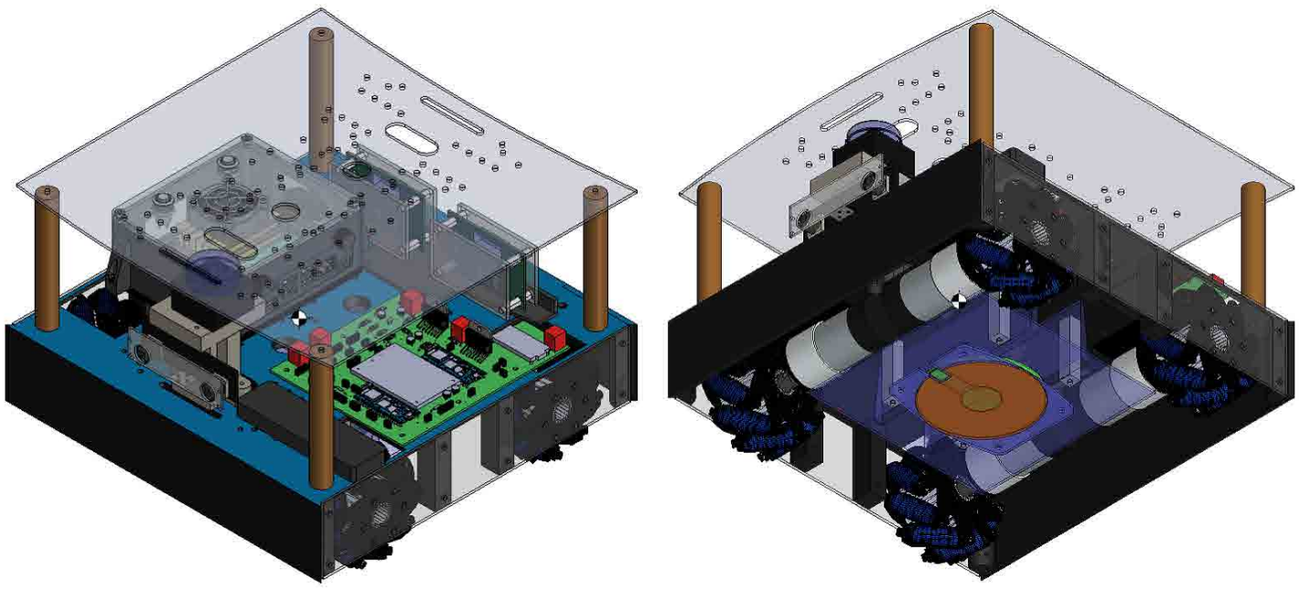
OmobotGuide

Shihab Uddin Ahamad



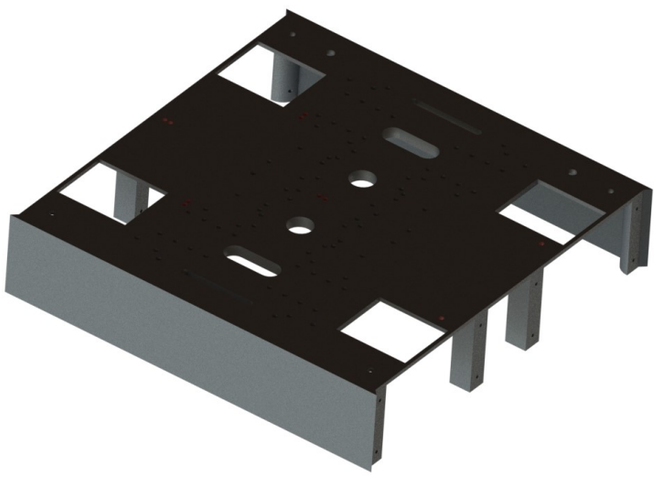
OMOBOT is a Ros-based autonomous **O**mnidirectional **MO**bile-ro**BOT** developed for periodically exploring indoor rooms, monitoring people, detecting anomalies (human falls), and sending an email or text to a set of designated responders in case of any anomaly detection. This document contains a complete walkthrough of Omobot's software integration with the hardware systems.

1 Hardwares and Bill of Material

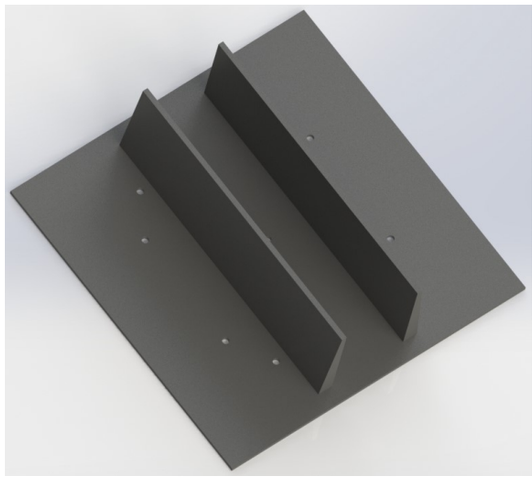
Table 1: Cost Breakdown

id	Components	Unit	Link	Cost (USD)
	Micro-Controller, Sensors, and Actuators			
1	Jetson Nano (Dev-4GB) and accessories	1	https://a.co/d/a8wsuNv	196
2	Arduino Nano RP2040 Connect	1	https://a.co/d/jhbh6Mf	29
3	Arduino Nano 33 IoT	1	https://a.co/d/5qcbEW4	27
4	IMX219-88 CSI Camera	2	https://a.co/d/0fRtf6h	56
5	Lidar (LD19)	1	https://a.co/d/3DeroNX	77
6	DC gear motor with encoder and wheels	4	https://a.co/d/6FwmeZt	80
	Electronics and Power System			
7	PCB Board (Dual Layer-Through Hole)	1	https://shorturl.at/cE139	3
8	Electronic Components Kit	1	https://a.co/d/8nNdC62	25
9	Motor Driver (LM298N)	2	https://a.co/d/iZR2ZE3	4
10	Lipo Battery (5200mAh)	1	https://a.co/d/c5rTGa3	27
11	Power supply - Buck-Boost Module	2	https://a.co/d/hLJqj2X	36
12	3Cell balance charging module with BMS	1	https://a.co/d/5NVqlxX	4
13	Wireless charging system (12V-2A DC)	1	https://a.co/d/53FImwY	25
	3D Printing, Connectors, and Assembly			
14	3D printed parts (approx 1KG)	1	https://a.co/d/gPwrX3i	13
15	Nuts and Bolts Assortment Kit	1	https://a.co/d/fmIZgGk	9
16	JST Connectors	1	https://a.co/d/cO7cxVQ	5
17	Ribbon Wires	1	https://a.co/d/0YIouvE	8
18	Header Connectors	1	https://a.co/d/iYn1X0y	2
19	Misc (glue, tape, veroboard)			5
	Total			636

2 3D Printed Parts



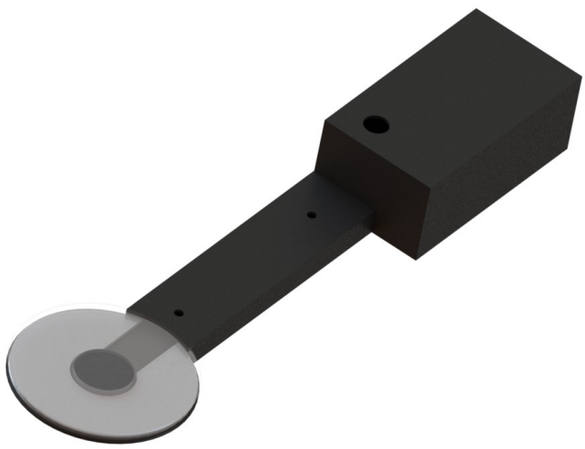
(a)



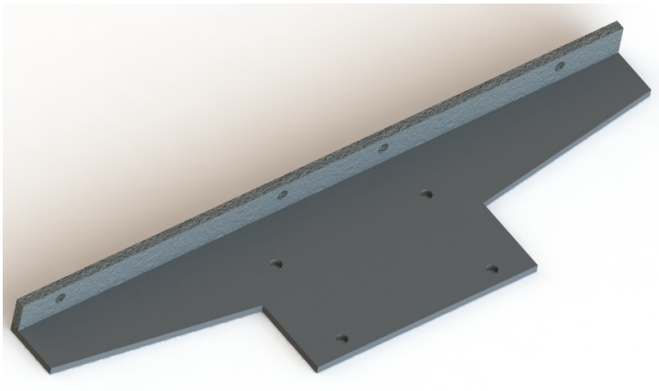
(b)



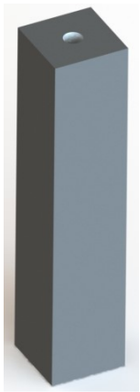
(c)



(d)



(e)



(f)

Fig. 1: Parts to be 3D printed; (a) Main Base, (b) Battery Mount, (c) Lidar/Camera Mount, (d) Wireless Charging Dock, (e) Buck/Boost Power Supply Mount, (f) Extender for Connecting Battery Mount with Main Base

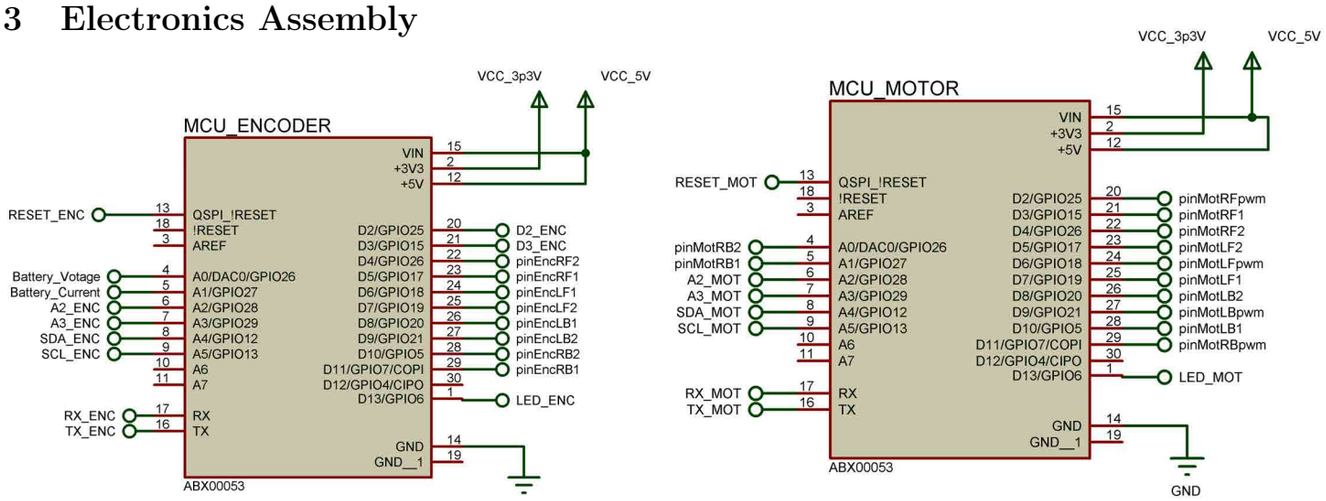
All the parts are designed in Solidworks and printed using PETG filament. The printing configuration is shown below:

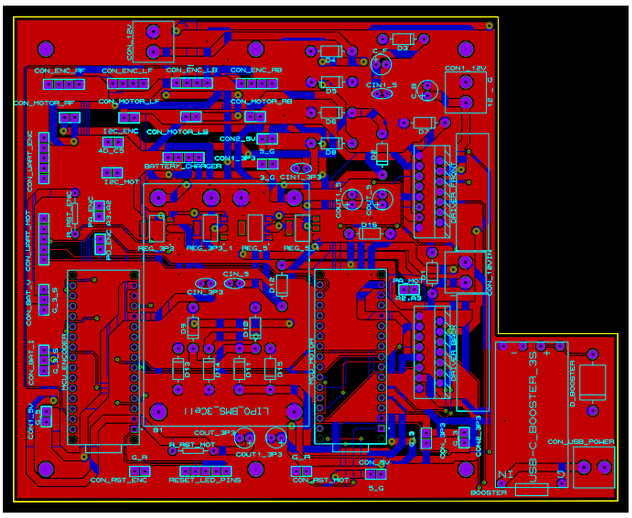
Table 2: Printer Setup

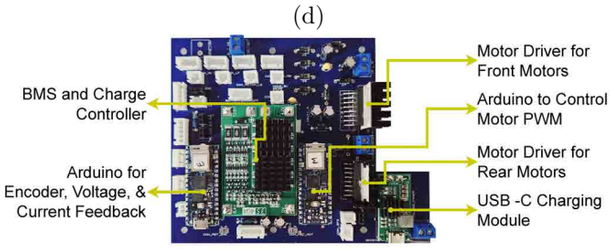
Printer	Filam Diam. (mm)	Nozzle Diam. (mm)	Bed Temp. (C)	Ext. Temp. (C)	Infill Density (%)	Layer Height (mm)
Creality CR-10S5	1.75	0.4	75	245	35	0.16

The step-by-step assembly is described in the Hardware Installation Guide

3 Electronics Assembly







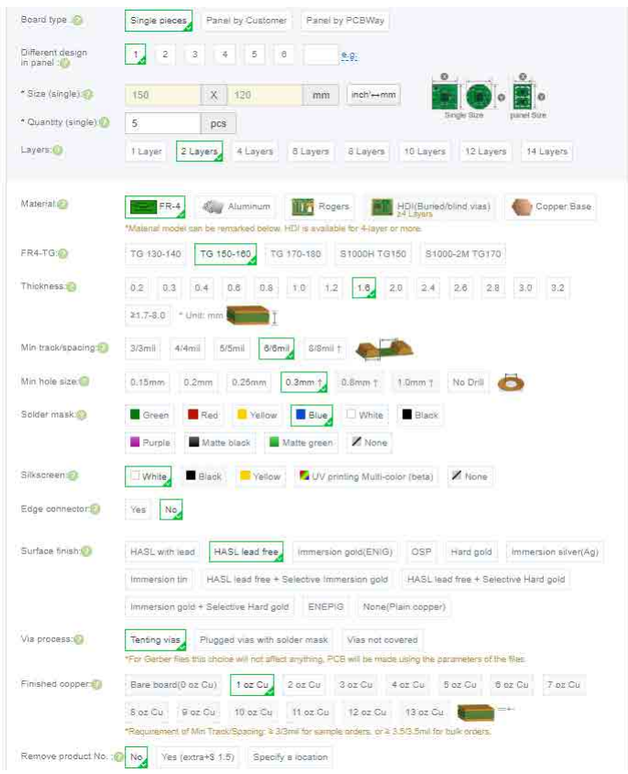


Fig. 2: PCB Schematics and Assembly; (a) Schematic of the Arduino-Enc, (b) Schematic of the Arduino-Mot, (c) Full Schematic Diagram, (d) PCB Layout, (e) Full PCB assembly, (f) PCB Printing Option

4 Hardware Installation Guide

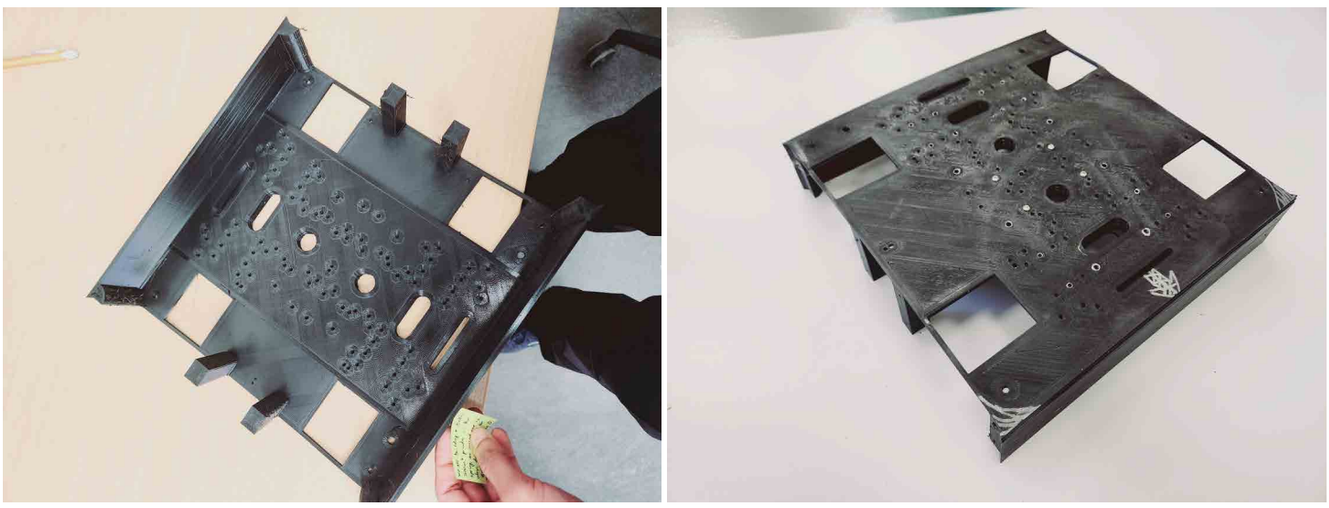
4.1 Get the Toolsets, Nut, and Bolts set ready



(a) (b)

Fig. 3: (a) Driver Tool-set, any tool-set with Allen key, Star and Flat screwdriver will work, (b) Multi-sized 3mm, 4mm, and 5mm bolts and nuts set (Table1-item15), 3mm bolt will mostly be used, you can buy separately instead of a set

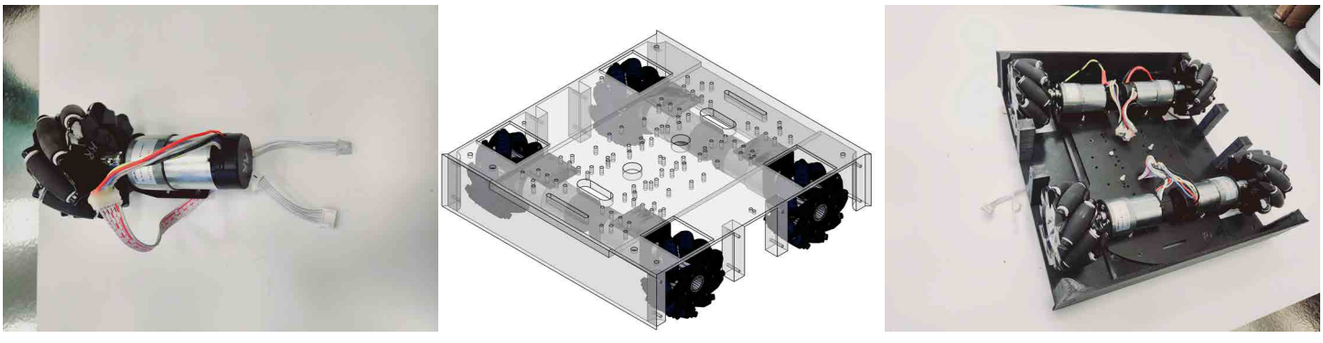
4.2 Print the Base



(a) (b)

Fig. 4: (a) 3D-printed-Base bottom view, STL file can be found here Base.STL, (b) Top view

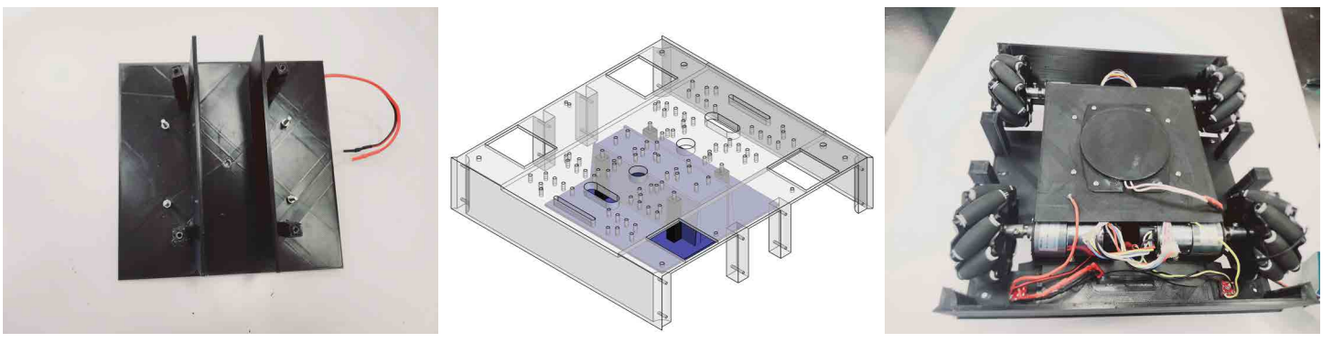
4.3 Assemble the four wheels



(a) (b) (c)

Fig. 5: (a) DC motor connection. divide the 6pins of the encoder into 4pins and 2pins, 2pins being the power pins which will be connected to CON_MOTOR_<Motor ID> as shown in (Figure **??** (*C*)), and the 4pins will be connected to the CON_ENC_<Motor ID> connectors, (b) Visual of mounting model (c) Bottom View after Assemble

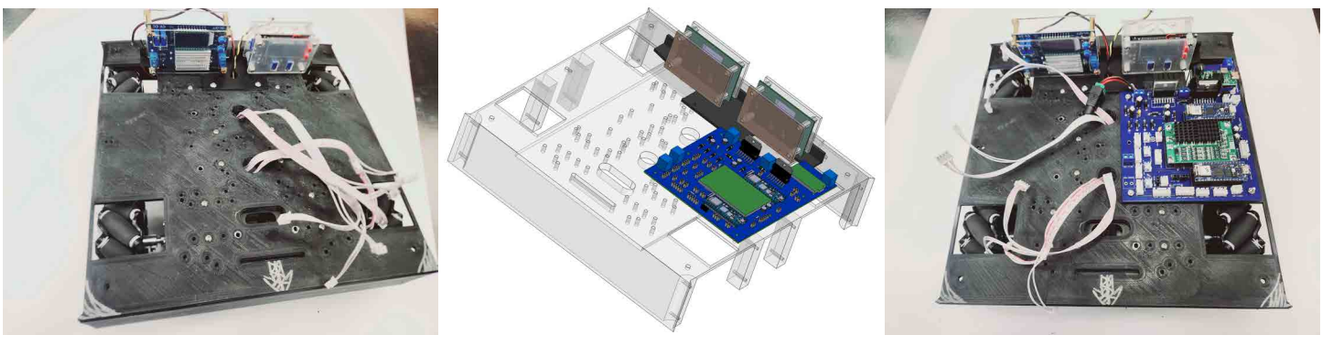
4.4 Assemble the Wireless Power Receiver and Battery Holder



(a) (b) (c)

Fig. 6: (a) Battery Holder, STL file can be found in BatteryHolder.STL, (b) Visualization of assembly, (c) Mounted at the bottom with the wireless receiver from (Table1-item13) on the 3D-printed holders WirelessChangerHolder-Bottom.STL and WirelessChangerHolderTop.STL and fix to the Base using the extender Extender3p47.STL and 3mm bolts

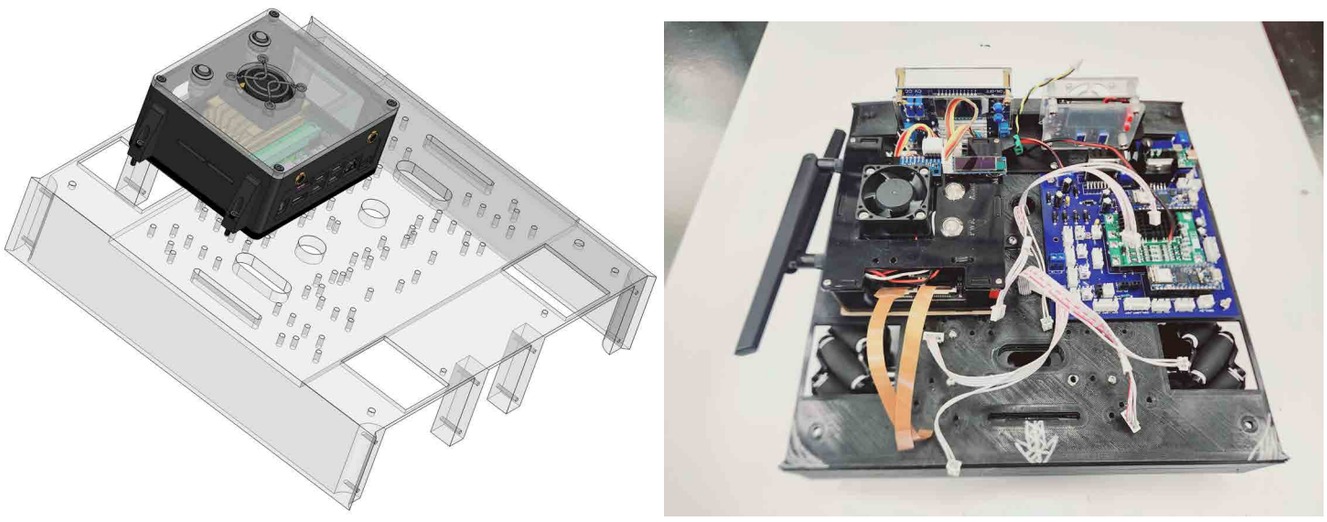
4.5 Assemble the Buck/Booster Power Supply and PCB



(a) (b) (c)

Fig. 7: (a) Mount the Buck-Booster modules from (Table1-item11) on the 3D printed BuckHolder.STL mounts, (b) Visualization of assembly, (c) Final assembly with PCB mounted

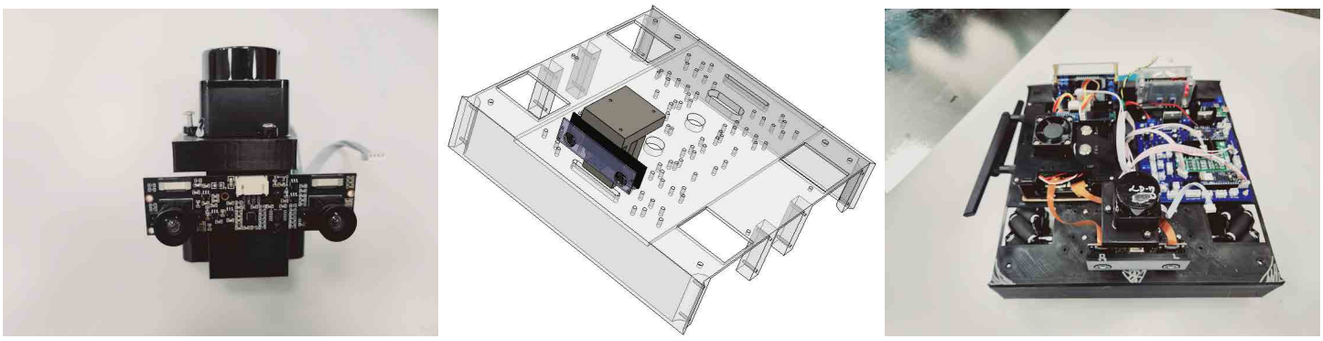
4.6 Assemble the Jetson Nano



(a) (b)

Fig. 8: (b) Visualization of assembly, (c) Mount the jetson Nano on the Base

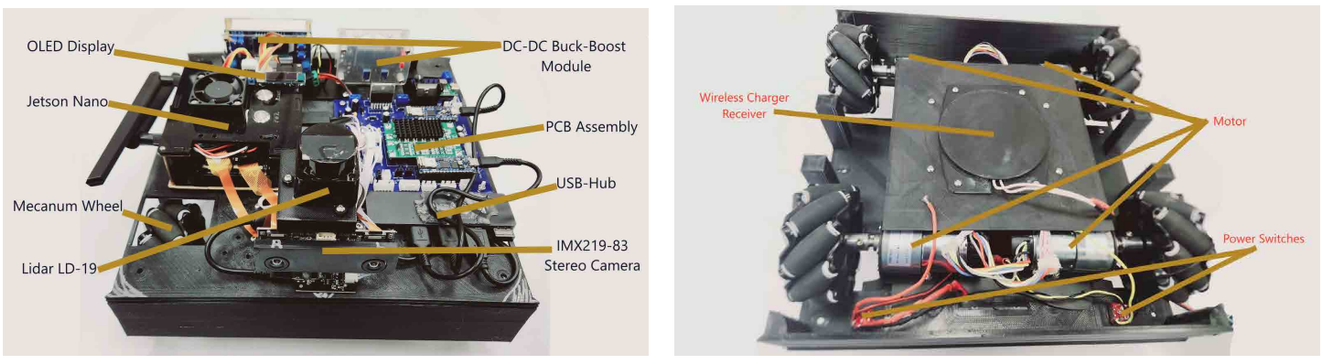
4.7 Assemble the Lidar and Stereo Camera



(a) (b) (c)

Fig. 9: (a) Mount the Lidar and Camera modules from (Table1-item4, item5) on the 3D printed LidarCamera-Holder.STL mounts, (b) Visualization of assembly, (c) Final assembly

4.8 Finale Assembly



(a) (b)

Fig. 10: (b) Top View of the Final assembly after all connection, (c) Bottom View of the Final assembly after all connection